

- ① 공분산·상관계수는 $E[XY]$ 부터 ② 조건부 다루면 전기댓값/총분산법칙 떠올리기 ③ 채점: origin 풀이 / /solve-problem

떠올릴 개념

- 공분산: $\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$; 상관계수 $\rho = \text{Cov}(X,Y)/(\sigma_X \sigma_Y) \in [-1,1]$
- 공분산 쌍선형성: $\text{Cov}(aX+bY, Z) = a\cdot\text{Cov}(X,Z)+b\cdot\text{Cov}(Y,Z)$; $\text{Var}(aX+bY)=a^2\text{Var}X+b^2\text{Var}Y+2ab\cdot\text{Cov}$
- 조건부기댓값: $E[X | Y=y] = \int x f_{\{X|Y\}}(x|y) dx$, 단 $f_{\{X|Y\}}(x|y)=f_{\{X,Y\}}(x,y)/f_Y(y)$
- 전기댓값정리(반복기댓값): $E[X] = E[E[X | Y]]$ (안쪽은 y 의 함수, 바깥쪽은 Y 로 평균)
- 총분산법칙: $\text{Var}(X) = E[\text{Var}(X | Y)] + \text{Var}(E[X | Y])$
- 랜덤합 $S = \sum_{i=1}^N X_i$ ($N \perp X_i$, iid): $E[S]=E[N]E[X]$, $\text{Var}(S)=E[N]\text{Var}(X)+\text{Var}(N)(E[X])^2$
- 결합정규: 선형결합도 정규 / 조건부도 정규 $X|Y=y \sim N(\mu_X+\rho(\sigma_X/\sigma_Y)(y-\mu_Y), (1-\rho^2)\sigma_X^2)$
- 모든 PMF/PDF는 명시 구간 밖에서 otherwise = 0

문항 목차 (총 5문항)

- [1] L08 · PSP 5.9.5 — [■] 결합정규 PDF에서 Cov·상관계수 ρ ·Var·독립성 — $\text{Cov}=E[XY]-E[X]E[Y]$
- [2] L08 · IPSRP 5.4.36 — [무] 결합정규 선형결합 $P(X+2Y \leq 3) \cdot \text{Cov}(X-Y, X+2Y)$ — 공분산 쌍선형성
- [3] L08 · PSP 7.5.6 — [■] 조건부PDF $f_{\{X|Y\}} \cdot E[X|Y=y]$ 정의대로 — 조건부기댓값 핵심 ★
- [4] L08 · PSP 7.5.7 — [■] 원판 균등 $f_{\{Y|X\}} \cdot E[Y|X=x]$ — 조건부기댓값(반복기댓값/총분산 연습)
- [5] L08 · IPSRP 5.4.38 — [무] 결합정규 $E[Y|X=3] \cdot \text{Var}(Y|X=2)$ ·조건부확률 — 조건부정규·조건부분

5.9.5■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = ce^{-(2x^2-4xy+4y^2)}.$$

- (a) What are $E[X]$ and $E[Y]$?
- (b) Find the correlation coefficient $\rho_{X,Y}$.
- (c) What are $\text{Var}[X]$ and $\text{Var}[Y]$?
- (d) What is the constant c ?
- (e) Are X and Y independent?

Problem 36

Let X and Y be jointly normal random variables with parameters $\mu_X = -1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = -\frac{1}{2}$.

- Find $P(X + 2Y \leq 3)$.
- Find $\text{Cov}(X - Y, X + 2Y)$.

— 풀이 —

7.5.6 ■ Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_Y(y)$?
- (b) What is $f_{X|Y}(x|y)$?
- (c) What is $E[X|Y = y]$?

7.5.7■ Over the circle $X^2 + Y^2 \leq r^2$, random variables X and Y have the uniform PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/(\pi r^2) & x^2 + y^2 \leq r^2, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{Y|X}(y|x)$?
- (b) What is $E[Y|X = x]$?

Problem 38

Let X and Y be jointly normal random variables with parameters $\mu_X = 2$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 9$, and $\rho = -\frac{1}{2}$.

- Find $E[Y|X = 3]$.
- Find $\text{Var}(Y|X = 2)$.
- Find $P(X + 2Y \leq 5|X + Y = 3)$.

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